■ Road Guardian AI

YOLOv8n Pothole Detection Model — Accuracy & Performance Report

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1. Model Overview

Property	Value
Architecture	YOLOv8 Nano (YOLOv8n)
Base Pre-trained Weights	yolov8n.pt (COCO)
Training Strategy	Transfer Learning (pretrained=True)
Training Epochs	30 (converged)
Input Resolution	640 × 640 pixels
Total Parameters	3,011,043
Computational Cost	8.2 GFLOPs
Layers	130
Training Device	CPU
Ultralytics Version	8.4.23
Training Completed	18 March 2026
Confidence Threshold	≥ 60% (live inference)
IoU Threshold (NMS)	0.70

2. Final Validation Metrics (Best Checkpoint — Epoch 29)

The table below shows the metrics recorded on the **held-out validation set** at the best checkpoint epoch. mAP50 is the primary accuracy metric used in the object-detection community and represents mean Average Precision at 50 % Intersection-over-Union (IoU) overlap.

Metric	Value	Interpretation
Precision	79.6 %	Of every predicted pothole, 79.6 % were correct (true positives)
Recall	67.3 %	Of all real potholes, the model successfully detected 67.3 %
mAP @ 50	**77.2 %**	Primary accuracy: avg precision across all IoU ≥ 0.50 thresholds
mAP @ 50–95	**49.8 %**	Strict accuracy: avg across IoU 0.50 → 0.95 (industry standard)
Val Box Loss	1.302	Bounding-box regression loss (lower = better localisation)

Val Cls Loss	1.083	Classification loss (lower = fewer misclassifications)
Val DFL Loss	1.301	Distribution Focal Loss for box coordinate refinement
Fitness Score	0.498	Composite weighted score used to select the best checkpoint

3. Training Progression (Epoch-by-Epoch Key Milestones)

Epoch	mAP@50	mAP@50-95	Precision	Recall	Note
1	25.2 %	12.9 %	48.6 %	24.4 %	Baseline — random init from pretrained
5	14.8 %	5.8 %	26.1 %	22.7 %	Warm-up instability
10	60.2 %	32.7 %	74.5 %	49.6 %	Rapid learning phase
15	73.8 %	43.6 %	77.1 %	63.3 %	Model stabilising
20	71.2 %	43.4 %	66.6 %	69.1 %	High recall, precision fluctuating
24	77.2 %	46.8 %	82.7 %	64.2 %	Near-peak performance
25	74.8 %	47.1 %	78.2 %	65.3 %	Slight dip — LR annealing
29	**77.2 %**	**49.8 %**	**79.6 %**	**67.3 %**	■ **BEST checkpoint saved**
30	77.5 %	49.8 %	83.0 %	66.1 %	Final epoch — marginal gain

Table 3 — Key epoch milestones extracted from the training checkpoint. Full 30-epoch data available in the training run logs.

4. Training Hyperparameters

Parameter	Value	Parameter	Value
Batch size	16	**Optimizer**	Auto (AdamW)
Learning rate (lr0)	0.01	**LR final (lrf)**	0.01
Momentum	0.937	**Weight decay**	0.0005
Warmup epochs	3	**Warmup momentum**	0.8
Mosaic augment	1.0 (on)	**Flip LR**	0.5
Scale	0.5	**Translate**	0.1
HSV Hue	0.015	**HSV Saturation**	0.7
HSV Value	0.4	**Erasing**	0.4
Box loss weight	7.5	**Cls loss weight**	0.5
DFL loss weight	1.5	**Close mosaic epoch**	10
AMP (mixed prec.)	True	**Seed**	0

5. Live Inference Configuration

Setting	Value	Effect
Model format	ONNX (OpenCV DNN)	Lightweight, no PyTorch needed at runtime
Fallback model	.pt via Ultralytics	Used if ONNX loading fails
Confidence threshold	≥ 60 %	Only high-confidence detections accepted
IoU NMS threshold	0.70	Non-max suppression aggressiveness
Accepted classes	pothole, damage, crack	Other classes are filtered out
Min bbox size	> 0 × 0 px	Degenerate boxes discarded
Image validation	≥ 100 × 100 px	Blurry/tiny uploads rejected
Deduplication	pHash + GPS proximity	Prevents duplicate hazard records

6. Recommendations for Accuracy Improvement

#	Recommendation	Expected Gain
1	Train for 50–100 epochs (model still improving at epoch 30)	+2–4 % mAP
2	Switch to YOLOv8s or YOLOv8m (larger backbone)	+5–10 % mAP
3	Train on GPU with larger batch size (32–64)	Faster + better augmentation
4	Add night/rain/wet-road images to dataset	Better real-world robustness
5	Use CutMix / MixUp augmentation (currently off)	+1–2 % mAP
6	Lower confidence threshold to 0.45 to increase recall	+8 % recall